

Which of the following catabolic pathways directly produce both ATP (or GTP) and NADH?

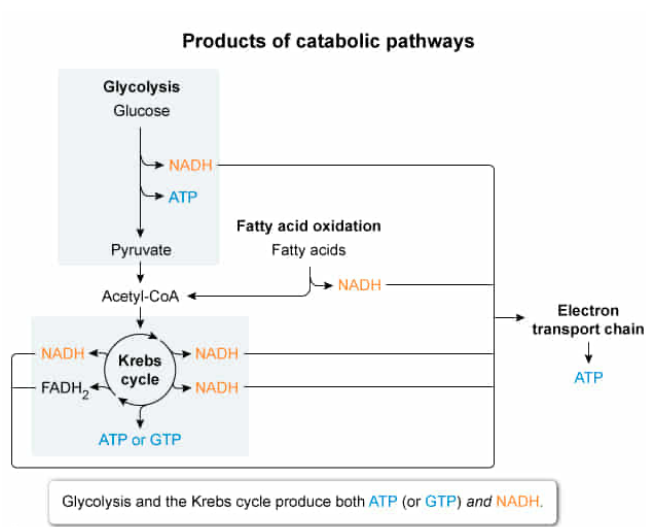
- I. Fatty acid oxidation
- II. Glycolysis
- III. The Krebs cycle
- IV. The electron transport chain

- ☐ A. I and II only [2%]
- ✓ ☒ B. II and III only [54%]
- ☐ C. I, II, and III only [22%]
- ☐ D. II, III, and IV only [20%]

Correct

54% answered correctly

Explanation:



Catabolic pathways oxidize biological fuel molecules (eg, glucose, fatty acids) to simpler molecules and use the released energy to synthesize ATP or energetically equivalent molecules, such as GTP. The ATP and GTP can then provide energy for **anabolic processes**, such as protein synthesis, or for processes that generate mechanical work, such as **muscle contractions**.

Some catabolic pathways synthesize ATP (or GTP) directly; others synthesize **reduced electron carriers** such as NADH, which can then enter the **electron transport chain** to produce ATP; and some pathways do both.

Glycolysis (Number II) produces 2 ATP molecules and 2 NADH molecules per glucose molecule consumed. The pyruvate molecules resulting from glycolysis

are then converted to acetyl-CoA. Acetyl-CoA can enter the **Krebs cycle (Number III)**, which produces 1 ATP molecule (or GTP, depending on the organism), 3 NADH molecules, and 1 FADH₂ molecule per acetyl-CoA molecule. Therefore, both glycolysis and the Krebs cycle produce ATP (or GTP) *and* NADH.

(Number I) **Fatty acid oxidation** directly produces NADH and acetyl-CoA. These molecules can then enter the electron transport chain and the Krebs cycle, respectively—*eventually* producing ATP—but fatty acid oxidation does *not* produce ATP on its own.

(Number IV) The electron transport chain produces ATP by *consuming* NADH (and FADH₂), not producing it.

Educational objective:

Catabolic processes oxidize biological fuel molecules, eventually producing ATP. Some processes produce ATP (or energetically equivalent molecules) directly, others produce reduced cofactors such as NADH, and some processes produce both.

Time spent:0 sec

Subject:
Biochemistry

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System:
Metabolic Reactions